

(MATH000) Mathematical Language

Notation and Conventions

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Part I. General Mathematical Language

1 General Conventions

These conventions apply throughout. Distinct typographic forms are assigned distinct roles, and clarity takes precedence over brevity. When multiple notations are listed for the same concept, the first is used throughout.

Item	Convention
Scalars	Ordinary mathematical italic letters, for example a, b, c, λ, t .
Vectors	Bold lowercase letters, for example $\mathbf{u}, \mathbf{v}, \mathbf{x}$. Unless otherwise stated, vectors are column vectors.
Matrices	Bold uppercase letters, for example $\mathbf{A}, \mathbf{B}, \mathbf{M}$.
Number systems	Blackboard bold, for example $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$.
Sets	Ordinary uppercase letters such as A, B, X, Y , unless a field-specific convention requires otherwise.
Families and collections	Calligraphic letters when useful, for example $\mathcal{F}, \mathcal{T}, \mathcal{B}$.
Named operators	Upright Roman type, for example $\det, \ker, \operatorname{im}, \operatorname{rk}$, and tr .
Large operators	Display size in displayed mathematics and notation tables, with limits above and below when appropriate.
Greek variants	Distinct glyphs are treated as distinct symbols; in particular, ε and ϵ are not interchangeable.

Notation	Meaning
$(a_1, \dots, a_n)$	Ordered tuple; parentheses are also the default matrix delimiters.
$[a, b]$	Closed interval.
$\{a_1, \dots, a_n\}$	Set delimiters.
$\{x \in A \mid P(x)\}$	Set-builder notation.
$\langle u, v \rangle$	Pairing or inner product, according to context.
$ x $	Absolute value, modulus, cardinality, or order, according to context.
$\ x\ $	Norm.
s.t.	Abbreviation for “such that” when naturally required.
$a = b$	Equality.
$a := b$	a is defined to be b .
$a \equiv b$	Identity, equivalence, or congruence, according to context.
$a \cong b$	Isomorphism or congruence, according to context.
$a \simeq b$	Context-dependent equivalence when standard in the area.
$a \approx b$	Approximation.
$f: X \rightarrow Y$	Function or map from X to Y .
$x \mapsto f(x)$	Elementwise rule of a map.
$g \circ f$	Composition: first f , then g .
$f _A, f \upharpoonright_A$	Restriction of f to A .
id_X	Identity map on X .
$\rightarrow$	Map, direction, or field-specific convergence arrow.
$\mapsto$	Elementwise assignment.
$\Rightarrow$	Logical implication.
$\Leftrightarrow$	Logical equivalence.

2 Theorem-like Environments

Item	Convention
Definition	Precise meaning of a term, object, property, or construction.
Theorem	Principal mathematical result.
Proposition	Useful result of more limited scope.
Lemma	Supporting result, usually for a later proof.
Corollary	Result following readily from a preceding result.
Conjecture	Statement proposed to be true but not proved in full generality.
Hypothesis	Named assumption or proposed statement.
Formula	Distinguished computational identity or closed form.
Algorithm	Explicit step-by-step procedure.
Fact	Verified statement not requiring theorem status.
Example	Illustration of a definition, construction, phenomenon, or result.
Counterexample	Example disproving a general assertion or showing a hypothesis is necessary.
Remark	Supplementary context, warning, or qualification.
Problem	Mathematical task intended to be solved.
Open Problem	Problem not known to be solved in the stated form.
Idea	Decisive conceptual insight.
Strategy	Broad problem-solving plan.
Tactic	Local step within a larger strategy.

Proof labels are written as *Proof* and solution labels as *Solution*, each followed by a period.

Notation	Meaning
□	Ending for ordinary proofs and solutions.
■	Ending for substantial, difficult, or particularly important proofs.
Q.E.D.	Ending for exceptionally difficult, significant, or paper-level proofs.

Item	Convention
Displayed mathematics	Remains part of the surrounding sentence and receives punctuation when grammar requires it.
Equation numbering	Number equations that are referenced later or benefit from a stable label.
Logical prose	Use words such as “and” and “or” unless symbolic logic improves clarity.
Abbreviations	Use established abbreviations only when standard and unambiguous.

Part II. Notation by Mathematical Area

3 Set Theory and Logic

Uppercase letters such as A, B, X, Y usually denote sets, while lowercase letters usually denote their elements.

Notation	Meaning
$\mathbb{N} = \{1, 2, 3, \dots\}$	Positive integers.
$\mathbb{N}_0 = \{0, 1, 2, 3, \dots\}$	Nonnegative integers.
$\mathbb{Z}$	Integers.
$\mathbb{Q}$	Rational numbers.
$\mathbb{R}$	Real numbers.
$\mathbb{C}$	Complex numbers.
$\mathbb{H}$	Quaternion division ring.
$\emptyset, \emptyset$	Empty set.
$x \in A$	x is an element of A .
$x \notin A$	x is not an element of A .
$A \subseteq B$	A is a subset of B ; equality is allowed.
$A \subsetneq B$	A is a proper subset of B .
$A \supseteq B$	A is a superset of B ; equality is allowed.
$A \supsetneq B$	A is a proper superset of B .
$A \cup B$	Union.
$A \cap B$	Intersection.
$A \setminus B, A - B$	Set difference.
A^c	Complement of A in the ambient set.
$\mathcal{P}(A), 2^A$	Power set of A .
$A \times B$	Cartesian product.
$A \sqcup B$	Disjoint union.
$ A , \#A$	Cardinality.
$\{x \in A \mid P(x)\}$	Elements $x \in A$ satisfying $P(x)$.
$x \sim y$	Equivalent elements under an understood equivalence relation.
$A/\sim$	Set of equivalence classes.
$\forall x \in A$	For every $x \in A$.
$\exists x \in A$	There exists $x \in A$.
$\exists! x \in A$	There exists a unique $x \in A$.
$\nexists x \in A$	There does not exist $x \in A$.
$P \Rightarrow Q$	P implies Q .
$P \nRightarrow Q$	P does not imply Q .
$P \Leftrightarrow Q$	P if and only if Q .
$P \wedge Q$	Logical conjunction.
$P \vee Q$	Logical disjunction.
$\neg P$	Logical negation.
$\text{dom } f$	Domain of f .
$\text{codom } f$	Codomain of f .
$f(A)$	Image of A under f .
$f^{-1}(B)$	Preimage of B under f .
$\text{im } f$	Image of the map f .
$f: X \hookrightarrow Y$	Injective map.

Notation	Meaning
$f: X \twoheadrightarrow Y$	Surjective map.
$f: X \xrightarrow{\sim} Y$	Isomorphism; for sets, a bijection.
$\mathbb{1}_A(x)$	Indicator function of A .
$\aleph_0$	Cardinality of a countably infinite set.
$\mathfrak{c}$	Cardinality of the continuum.

4 Calculus and Analysis

The differential d is upright. The glyphs ε and ϵ are distinct symbols; ε is the default small positive quantity in analysis.

Notation	Meaning
$f'(a)$	First derivative of f at a .
$f^{(n)}(a)$	n th derivative of f at a .
$\frac{\mathrm{d}y}{\mathrm{d}x}$	Derivative of y with respect to x ; d is upright.
$\frac{\partial f}{\partial x_i}, \partial_i f$	Partial derivative with respect to x_i .
$\int_a^b f(x) \, \mathrm{d}x$	Definite integral over $[a, b]$.
$\iint_D f \, \mathrm{d}A$	Double integral over D .
$\iiint_\Omega f \, \mathrm{d}V$	Triple integral over Ω .
$\oint_C f(z) \, \mathrm{d}z$	Contour integral over a closed curve C .
$\sum_{k=1}^n a_k$	Finite sum.
$\sum_{k=1}^{\infty} a_k$	Infinite series.
$\prod_{k=1}^n a_k$	Finite product.
$\lim_{x \rightarrow a} f(x)$	Limit as x approaches a .
$\ln x, \log_e x$	Logarithm to base e .
$\lg x, \log_{10} x$	Logarithm to base 10.
$\mathrm{lb} x, \log_2 x$	Logarithm to base 2.
$\log_a x$	Logarithm to base a .
$\arcsin x$	Principal inverse sine.
$\arccos x$	Principal inverse cosine.
$\arctan x$	Principal inverse tangent.
$\operatorname{arcsec} x$	Principal inverse secant.
$\operatorname{arccsc} x$	Principal inverse cosecant.
$\operatorname{arccot} x$	Principal inverse cotangent.
$\Gamma(z)$	Gamma function.
$B(x, y)$	Beta function.
$\zeta(s)$	Riemann zeta function unless another zeta function is defined.
∇f	Gradient of f .

Notation	Meaning
$\nabla \cdot \mathbf{F}$	Divergence of $\mathbf{F}$.
$\nabla \times \mathbf{F}$	Curl of $\mathbf{F}$.
Δf	Laplacian of f .
$D\mathbf{F}(\mathbf{a})$	Total derivative, represented by the Jacobian matrix in coordinates.
$\text{Jac}(\mathbf{F})$	Jacobian determinant.
$\text{Hess } f$	Hessian matrix of f .
$\mathbf{r}(t)$	Parametrized curve.
ds	Element of arc length.
$\int_C \mathbf{F} \cdot d\mathbf{r}$	Line integral along C .
$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$	Flux integral through S .
$x_n \rightarrow x$	Convergence of a sequence.
$a_n \nearrow L$	Monotone increasing convergence to L .
$a_n \searrow L$	Monotone decreasing convergence to L .
$\limsup a_n$	Limit superior.
$\liminf a_n$	Limit inferior.
$f_n \rightarrow f$	Pointwise convergence.
$f_n \rightrightarrows f$	Uniform convergence.
$f_n \xrightarrow{\text{a.e.}} f$	Convergence almost everywhere.
$\ f\ _\infty$	Supremum norm.
$C^k(I)$	k times continuously differentiable functions on I .
$C^\infty(I)$	Smooth functions on I .
$B_r(x)$	Open ball of radius r centered at x .
$\overline{B}_r(x)$	Closed ball of radius r centered at x .
$z = x + iy$	Complex number with real part x and imaginary part y .
$\text{Re } z$	Real part of z .
$\text{Im } z$	Imaginary part of z ; distinct from lowercase im for image.
$\bar{z}$	Complex conjugate.
$ z $	Complex modulus.
$\arg z$	Argument of z .
$\text{Res}(f; a), \text{Res}_{z=a} f(z)$	Residue of f at a .

5 Linear Algebra

Notation	Meaning
$\mathbf{A} = (a_{ij})$	Matrix with (i, j) -entry a_{ij} .
$\mathbf{A}^\top$	Transpose of $\mathbf{A}$.
$\mathbf{A}^H$	Conjugate transpose of $\mathbf{A}$.
$\mathbf{A}^{-1}$	Inverse of $\mathbf{A}$.
$\mathbf{I}_n$	$n \times n$ identity matrix.
$\mathbf{0}$	Zero vector or zero matrix, with dimension understood.
$\mathbf{D}$	Diagonal matrix when a local name is required.
$\text{diag}(d_1, \dots, d_n)$	Diagonal matrix with entries $d_1, \dots, d_n$.
$\mathbf{1}$	All-ones vector.
$\det(\mathbf{A}), \mathbf{A} $	Determinant.

Notation	Meaning
$\text{rk}(\mathbf{A}), \text{rank}(\mathbf{A})$	Rank.
$\text{tr}(\mathbf{A})$	Trace.
$\text{adj}(\mathbf{A})$	Adjugate matrix.
$\text{Col}(\mathbf{A})$	Column space.
$\text{Row}(\mathbf{A})$	Row space.
$\text{Null}(\mathbf{A}), \ker \mathbf{A}$	Null space.
$\ker T$	Kernel of a linear map T .
$\text{im } T$	Image of a linear map T .
$\mathbf{e}_i$	i th standard basis vector.
$\dim V$	Dimension of V .
$\text{Span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$	Span of the listed vectors.
$U \oplus V$	Direct sum.
$\langle \mathbf{u}, \mathbf{v} \rangle$	Inner product.
$\ \mathbf{v}\ $	Norm of $\mathbf{v}$.
$\mathbf{u} \perp \mathbf{v}$	Orthogonality.
$\mathbf{u} \times \mathbf{v}$	Cross product in $\mathbb{R}^3$.
$p_{\mathbf{A}}(\lambda), \chi_{\mathbf{A}}(\lambda)$	Characteristic polynomial of $\mathbf{A}$.
$m_{\mathbf{A}}(\lambda), \mu_{\mathbf{A}}(\lambda)$	Minimal polynomial of $\mathbf{A}$.
$\rho(\mathbf{A})$	Spectral radius.
$\sigma_i(\mathbf{A})$	i th singular value of $\mathbf{A}$.

6 Differential Equations

Notation	Meaning
y'	First derivative with respect to the understood independent variable.
y''	Second derivative.
$\dot{x}$	First derivative of x with respect to time.
$\ddot{x}$	Second derivative of x with respect to time.
y_h, y_c	Homogeneous or complementary part of a solution.
y_p	Particular solution.
$c_1, c_2, \dots$	Arbitrary constants in a general solution.
$W(y_1, y_2)(t), W[y_1, y_2](t)$	Wronskian of y_1 and y_2 .
$\mathcal{L}\{f(t)\}(s)$	Laplace transform of f .
$\hat{f}(\omega), \mathcal{F}f(\omega)$	Fourier transform.

7 Combinatorics and Discrete Mathematics

Notation	Meaning
$\binom{n}{r}, {}_nC_r$	Number of r -element subsets of an n -element set.
$\left(\binom{n}{r}\right), {}_nH_r$	Number of r -combinations with repetition from n types; equals $\binom{n+r-1}{r}$.
$n!$	Factorial.

Notation	Meaning
$n^{\underline{k}}, (n)_k$	Falling factorial.
$n^{\overline{k}}, n^{(k)}$	Rising factorial.
$O(g(n))$	Big-O upper asymptotic bound.
$o(g(n))$	Little-o strict asymptotic bound.
$\Theta(g(n))$	Two-sided asymptotic order.
$\Omega(g(n))$	Asymptotic lower bound.
$G = (V, E)$	Graph with vertex set V and edge set E .
$V(G)$	Vertex set of G .
$E(G)$	Edge set of G .
$N(v), \Gamma(v)$	Neighborhood of vertex v .
$\deg(v)$	Degree of vertex v .
K_n	Complete graph on n vertices.
P_n	Path graph on n vertices.
C_n	Cycle graph on n vertices.

8 Probability and Statistics

Probability and expectation use blackboard-bold operators.

Notation	Meaning
$\mathbb{P}(A), P(A)$	Probability of event A .
$\mathbb{P}(A \mid B), P(A \mid B)$	Conditional probability of A given B .
$\mathbb{E}[X], E[X]$	Expectation of random variable X .
$\text{Var}(X)$	Variance of X .
$\text{Cov}(X, Y)$	Covariance of X and Y .
$X \sim \text{Bin}(n, p)$	Binomial distribution.
$X \sim \text{Pois}(\lambda)$	Poisson distribution.
$X \sim \mathcal{N}(\mu, \sigma^2)$	Normal distribution with mean μ and variance σ^2 .
$X_n \xrightarrow{\text{a.s.}} X$	Almost-sure convergence.
$X_n \xrightarrow{\mathbb{P}} X$	Convergence in probability.
$X_n \xrightarrow{L^p} X$	Convergence in L^p .
$X_n \xrightarrow{d} X, X_n \Rightarrow X$	Convergence in distribution.

9 Geometry

Uppercase letters such as A, B, C denote points. In triangle notation, a, b, c denote the side lengths opposite A, B, C , respectively.

Notation	Meaning
$\triangle ABC$	Triangle with vertices A, B, C .
$\angle ABC$	Angle with vertex B .
$\overline{AB}$	Segment joining A and B .
AB	Length of $\overline{AB}$.
$[ABC]$	Area of $\triangle ABC$.
$\text{length}(C), \ell(C)$	Length of C .
$\text{area}(S)$	Area of S .
$\text{vol}(S)$	Volume of S .

Notation	Meaning
$\text{dist}(P, S)$	Distance from P to S .
$\text{conv}(S)$	Convex hull of S .
$\mathbb{E}^n$	n -dimensional Euclidean space.
$\mathbb{H}^n$	n -dimensional hyperbolic space.
$\parallel$	Parallel.
$\perp$	Perpendicular or orthogonal.

10 Number Theory

Unless otherwise stated, divisibility and congruence statements are made in $\mathbb{Z}$.

Notation	Meaning
$a \mid b$	a divides b .
$a \nmid b$	a does not divide b .
$\text{gcd}(a, b)$	Greatest common divisor, taken nonnegative.
$\text{lcm}(a, b)$	Least common multiple.
$a \equiv b \pmod{n}$	a is congruent to b modulo n .
$[a]_n$	Residue class of a modulo n .
$\mathbb{Z}/n\mathbb{Z}$	Ring of residue classes modulo n .
$(\mathbb{Z}/n\mathbb{Z})^\times$	Multiplicative group of units modulo n .
$a^{-1} \pmod{n}$	Multiplicative inverse of a modulo n .
$\left(\frac{a}{p}\right)$	Legendre symbol.
$\varphi(n)$	Euler totient function.
$\text{ord}_n(a)$	Multiplicative order of a modulo n .
$v_p(n)$	p -adic valuation of n .
$\tau(n)$	Number of positive divisors of n .
$\sigma(n)$	Sum of positive divisors of n .
$\mu(n)$	Möbius function.
$\text{rad}(n)$	Product of the distinct prime divisors of n .
$\pi(x)$	Number of primes not exceeding x .
$\mathbb{F}_q, \text{GF}(q)$	Finite field with q elements.

11 Algebra

Capital letters such as G, H denote groups, R, S rings, and F, K, L fields when no conflicting local notation is established.

Notation	Meaning
G	Group.
$e, 1_G$	Identity element of a group.
$ G $	Order of a finite group.
$\langle S \rangle$	Subgroup generated by S .
$H \leq G$	H is a subgroup of G .
$H \trianglelefteq G$	H is a normal subgroup of G .
G/H	Quotient group.
$\ker \varphi$	Kernel of φ .
$\text{im } \varphi$	Image of φ .

Notation	Meaning
$\text{Aut}(G)$	Automorphism group of G .
$Z(G)$	Center of G .
$C_G(x)$	Centralizer of x in G .
$\text{GL}_n(F)$	General linear group over F .
$\text{SL}_n(F)$	Special linear group over F .
$S_n, \mathfrak{S}_n$	Symmetric group on n symbols.
$A_n, \mathfrak{A}_n$	Alternating group on n symbols.
R	Ring.
$I \trianglelefteq R$	I is an ideal of R .
R/I	Quotient ring.
F	Base field in an algebraic context.
K/F	Field extension of F by K .
$F[x]$	Polynomial ring in x over F .
$\deg f$	Degree of a polynomial f .
$\text{char } F$	Characteristic of F .
$\text{Gal}(L/K)$	Galois group of L/K .
$\text{trdeg}_K L$	Transcendence degree of L over K .

12 Topology

Notation	Meaning
$(X, \mathcal{T})$	Topological space X with topology $\mathcal{T}$.
$A^\circ, \text{int } A$	Interior of A .
$\bar{A}, \text{cl } A$	Closure of A .
∂A	Boundary of A .
$\pi_1(X, x_0)$	Fundamental group of X based at x_0 .
$\chi(X)$	Euler characteristic of X .
$X \cong Y$	Homeomorphic spaces.
$X \simeq Y$	Homotopy equivalent spaces.

13 Miscellaneous Notation

Notation	Meaning
$\lfloor x \rfloor$	Floor of x .
$\lceil x \rceil$	Ceiling of x .
$\{x\}$	Fractional part $x - \lfloor x \rfloor$ when unambiguous.
$\text{sgn}(x)$	Sign of a real number or permutation, according to context.
Δq	Finite change in a quantity q .
ϵ_{ijk}	Levi-Civita symbol; ϵ is distinct from analysis ε .